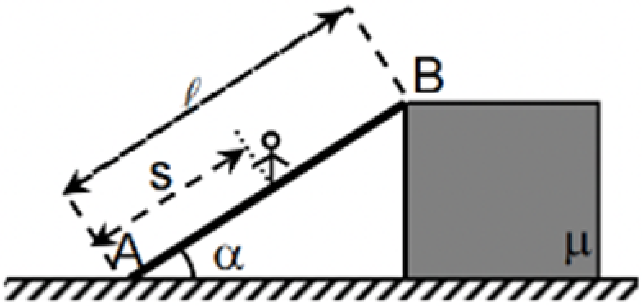
Q1

A thin rod of mass m_2 and length ℓ is leaning on a block with a mass m_1 at angle α with the horizontal (see figure). What is maximum distance S from A upto which a person of mass M at point A can go on the rod keeping block in equilibrium. It is given that rod is fixed at point A on ground. The coefficient of friction between block and ground is μ.



(1) $S=\left(\frac{\mu m_1}{\sin \alpha-\mu \cos \alpha}-m_2 \cos \alpha\right) \frac{\ell}{M \cos \alpha}$
(2) $S=\left(\frac{\mu m_1}{\sin \alpha+\mu \cos \alpha}-\frac{m_2 \cos \alpha}{2}\right) \frac{\ell}{M \cos \alpha}$
(3) $S=\left(\frac{\mu m_1}{\sin \alpha-\mu \cos \alpha}-\frac{m_2 \cos \alpha}{2}\right) \frac{\ell}{M \cos \alpha}$
(4) $S=\left(\frac{\mu m_1}{\sin \alpha+\mu \cos \alpha}-m_2 \tan \alpha\right) \frac{\ell}{M \cos \alpha}$

Q2

When the voltage applied to an X-ray tube increased from $V_1=15.5\mathrm{kV}$ to $V_2=31\ kV$, the wavelength interval between the K_α line and the cut-off wavelength of the continuous X-ray spectrum increases by a factor of 1.3. If the atomic number of the element of the target is z. Then the value of $\frac{z}{13}$ will be : (Take : hc $=1240 \mathrm{eVnm}$ and $\mathrm{R}=1 \times 10^7 / \mathrm{m})$

(1) 24

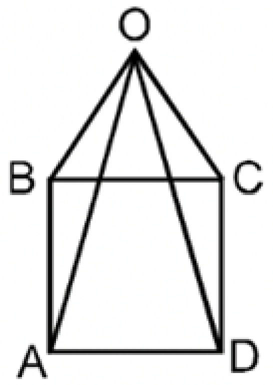
(2) 26

(3) 32

(4) 16

Q3

Eight identical resistances each 15Ω are connected along the edge of a pyramid having square base as shown. Find the equivalent resistance between A and D in Ω.



(1) 8

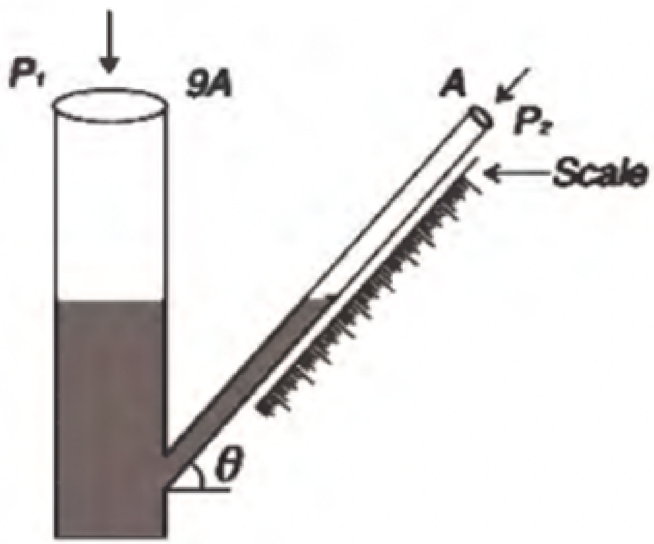
(2) 16

(3) 12

(4) 9

Q4

A manometer has a vertical of cross-sectional area $9 \mathrm{~A}$ and an inclined arm having area of cross section A. The density of the manometer liquid has a specific gravity of 0.74. The scale attached to the inclined arm can read up to $\pm 0.5 \mathrm{~mm}$. It is desired that the manometer shall record pressure difference $\left(P_1-P_2\right)$ up to an accuracy of $\pm 0.09 \mathrm{~mm}$ of water. To achieve this, the inclination angle θ of the inclined arm should be



(1) $\sin^{-1}(0.13)$

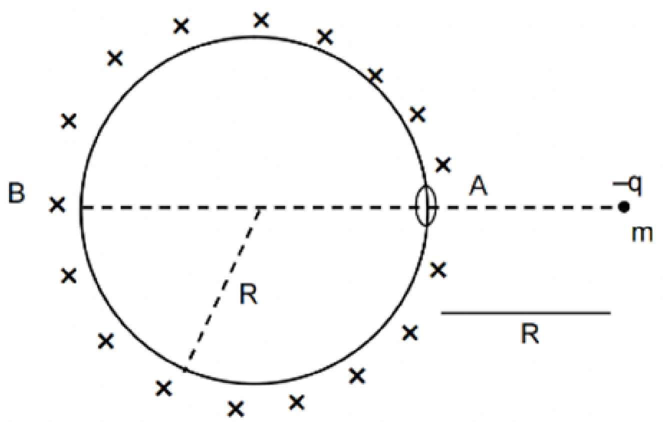
(2) $\sin^{-1}(0.03)$

(3) $\sin^{-1}(0.33)$

(4) $\sin^{-1}(0.23)$

Q5

Consider a uniformly charged non conducting, uniform spherical shell of radius R. Net charge on the shell is Q. There is a small hole in the shell at 'A' shown in the figure. A point charge $-q$ is released at a distance 2R from the centre as shown in the figure. The shell and point charge have same mass m. After release the point charge will move toward shell, passes through the hole and hit the shell at 'B'. Choose the correct option(s):



(1) The distance travelled by point charge till it hit the shell at B is $\frac{5R}{2}$

(2) Speed of the point charge when it hit the shell is $\sqrt{\frac{\mathrm{kQq}}{2\mathrm{mR}}}$

(3) Time taken by the point charge to travel from A to B is $\sqrt{\frac{8mR^3}{kQq}}$

(4) The distance travelled by the shell with constant velocity is R

Q6

A radio active nuclei A decays in B with decay constant λ. The nuclear B may further decay in stable nuclei C or D with decay constant λ and 2λ respectively. Initially at $t=0$, the number of nuclei of A is N_0 and number of all other nuclei is zero. If N_1, N_2, N_3 and N_4, are number of nuclei of A, B, C and D respectively at any time t. Then select correct relations.

(1) $\mathrm{N}_1 = \mathrm{N}_0 \mathrm{e}^{-\lambda t}$

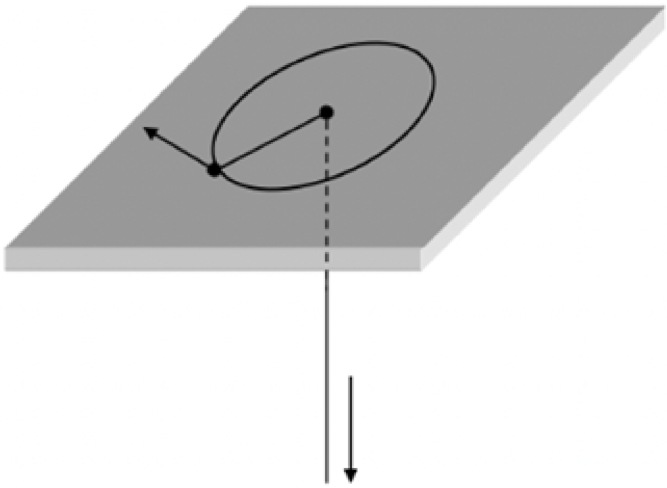
(2) $\mathrm{N}_2 = \frac{\mathrm{N}_0}{2}\left(\mathrm{e}^{-\lambda t} - \mathrm{e}^{-3\lambda t}\right)$

(3) $N_3 = \frac{1}{2}(N_0 - N_1 - N_2)$

(4) N4 = N1

Q7

A mass is attached to the end of a string. The mass moves on a friction less table, the string passes through a hole in the table, under which some one is pulling on the string to make it taut at all times. Initially, the mass moves in a circle of radius R with kinetic energy E_0. The string is slowly pulled, until the radius of the circle is halved. Choose the correct option(s)



(1) The tension in string is $\frac{2\mathrm{E}_0}{\mathrm{R}}$

(2) The tension in string is $\frac{\mathrm{L}^2}{\mathrm{mR}^3}$. Where L is angular momentum of the particle about center.

(3) Work done to reduce the radius half is $\frac{3L^2}{2m^2}$ Where L is angular momentum about center.

(4) Work done to reduce the radius half is $3E_0$

Q8

Two spherical soap bubbles in vaccum are connected through a narrow tube. Radius of left bubble is R and that of other is slightly smaller than R. Air flows from right to left very slowly. At any instant r_1, A_1, V_1, n_1 are radius, surface area, volume and number of moles of gas in the left bubble and $\mathrm{r}_2,\ \mathrm{A}_2,\ \mathrm{V}_2, \mathrm{n}_2$ are same for right bubble. Assume that temperature remains constant :

Suppose at any instant number of moles in left bubble is 4 times of number of moles in right bubble then select correct statement(s) :

(1) $r_2 = R\sqrt{\frac{2}{5}}$
(2) $r_1 = R\sqrt{\frac{8}{5}}$
(3) $A_1 + A_2 =$ constant
(4) $V_1 + V_2 =$ constant

Q9

An air column in a pipe, which is closed at one end will be in resonance with a vibrating tuning fork of frequency 264 Hz, if the length of the pipe (in cm) is: (Speed of sound in air $= 330 \mathrm{~m/s}$)

(1) 31.25

(2) 62.5

(3) 93.75

(4) 125

Q10

A parallel-plate capacitor is connected to a cell. Its positive plate A and its negative plate B have charges $+Q$ and $-Q$ respectively. A third plate C, identical to A and B, with charge $+Q$, is now introduced midway between A and B, parallel to them. Which of the following is/are correct?

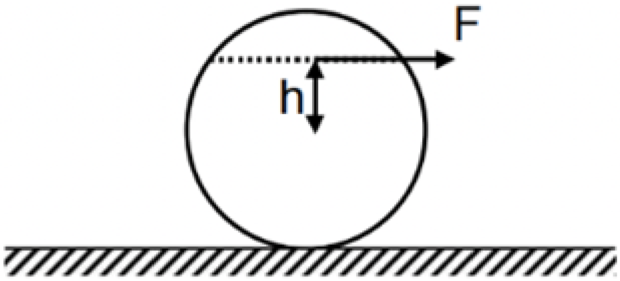
(1) The charge on the inner face of B is now $-\frac{3Q}{2}$.
(2) There is no change in the potential difference between A and B.

(3) The potential difference between A and C is one-third of the potential difference between B and C.

(4) The charge on the inner face of A is now $\frac{\mathrm{Q}}{2}$.

Q11

A solid sphere of mass m and radius R is placed on a rough horizontal surface. A horizontal force F is applied to sphere at a height $h, (0 \leq \mathrm{h} \leq \mathrm{R})$ from centre. If sphere rolls without slipping then,



(1) The maximum possible acceleration of sphere is $\frac{F}{M}$

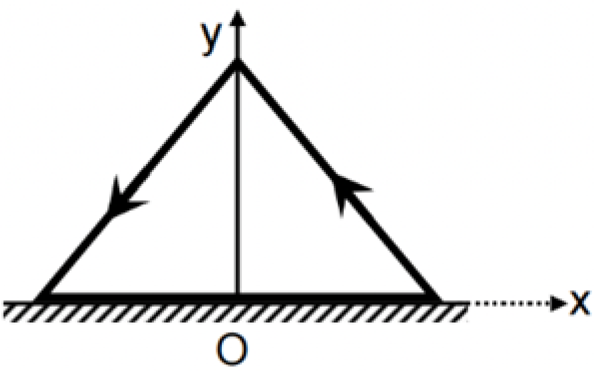
(2) The maximum possible acceleration of sphere is $\frac{10 \mathrm{~F}}{7 \mathrm{M}}$

(3) The magnitude of friction force acting on the sphere is zero if $h=\frac{2 R}{5}$.

(4) The magnitude of friction force acting on the sphere is zero if $h=\frac{3 R}{4}$.

Q12

A conducting frame in the shape of an equilateral triangle (mass m, side a) carrying a current I is placed vertically and a horizontal rough surface (coefficient of friction is μ). The frame is free to rotate about y-axis only. A magnetic field exists such that $\vec{B}=-B_0 y \hat{\imath}$. Then



(1) The maximum value of B_0 so that the frame does not rotate $\frac{2 \mu \mathrm{mg}}{\mathrm{Ia}^2}$

(2) The maximum value of B_0 so that the frame does not rotate $\frac{\mu \mathrm{mg}}{\mathrm{Ia}^2}$

(3) If seen from the top, the frame will have a tendency to rotate counter clockwise.

(4) If seen from the top, the frame will have a tendency to rotate clockwise.

Q13

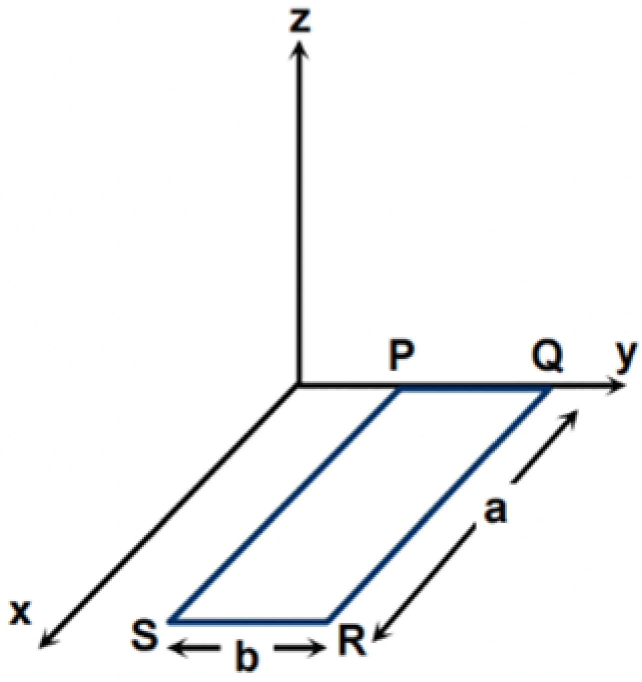
Consider a cell of emf E and internal resistance r. The current drawn (i) from cell can be varied. Output power from cell is same when current drawn is $\mathrm{i}_1 = 2\ \mathrm{A}$ and $\mathrm{i}_2 = 8\ \mathrm{A}$. For what value of current the output power will be maximum.

Q14

Two radioactive samples A_1 and A_2 having half life 3 years and 2 years respectively have been decaying for many years. Today the number of atoms in the sample A_1 is twice the number of atoms in the sample A_2. Both the samples had same number of atoms X years ago, calculate X.

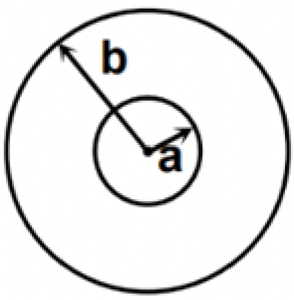
Q15

A rectangular loop PQRS made from a uniform wire has length a, width b and mass m. It is free to rotate about the arm PQ, which remains hinged along a horizontal line taken as the y-axis (see figure). Take the vertically upward direction as the z-axis. A uniform magnetic field $\vec{B} = (3\hat{i} + 4\hat{k})B_0$ exists in the region. The loop is held in the $x - y$ plane and a current I is passed through it. The loop is now released and is found to stay in the horizontal position in equilibrium. Find the ratio $\frac{\mathrm{mg}}{\mathrm{IB_0\ b}}$



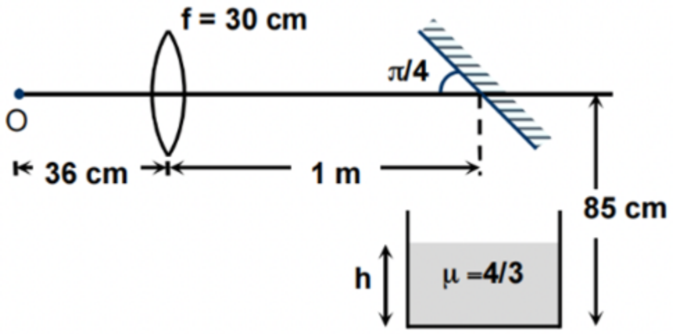
Q16

The current in the outer coil is varying with time as $I = 2t^2$. If the resistance of the inner coil is R and $b >> \mathrm{a}$ then the heat developed in the inner coil between $t = 0$ and t seconds is $\frac{4\mu_0^2\pi^2a^4t^3}{kb^2R}$. Find k.



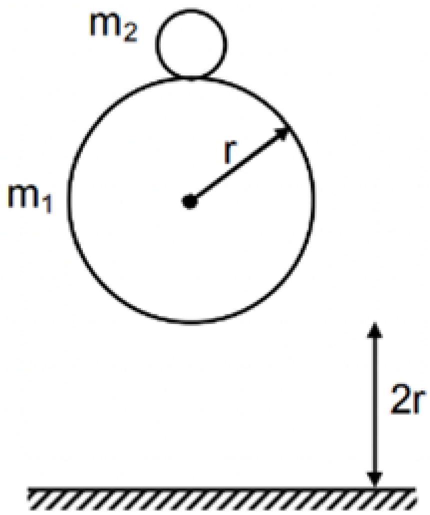
Q17

The image of the object O, shown in the figure, is formed at the bottom of the tank filled with water. If the water level in the tank is $h = 5k$, find k.



Q18

A small ball of mass m_2 rests on the top of a bigger ball of mass m_1 $(>> m_2)$. The centre of bigger ball is at height 3r above ground as shown. The balls are dropped from this situation. If the smaller ball bounces to a height 4nr from ground after its first collision with bigger ball. Find n. (Assume all collisions are elastic)



Q19

When heated above 918°C, iron changes its crystal structure from bcc to ccp structure without any change in the radius of atom. The ratio of density of the crystal before heating to its density after heating is:

(1) 1.890

(2) 0.918

(3) 0.725

(4) 1.231

Q20

Which of the following reactions will give primary alcohol as major product?

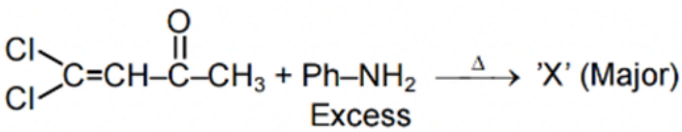
(1) Propene $\xrightarrow[2.\ H_2O_2/OH^-]{1.\ B_2H_6/THF}$

(2) Propyne $\xrightarrow{HgSO_4/dil.H_2SO_4}$

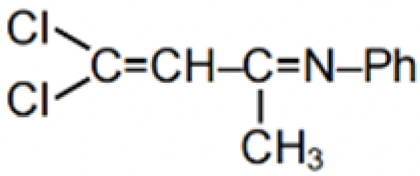
(3) Propene $\xrightarrow{dilute\ \ H_2SO_4}$

(4) Acetyl Chloride $\xrightarrow{H_2/Pd/BaSO_4}$

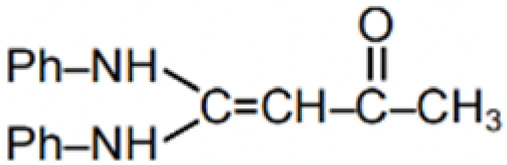
Q21

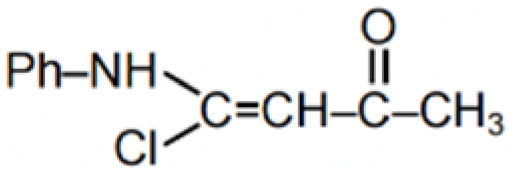


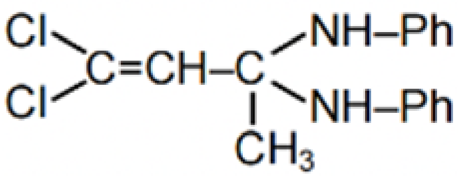
X will be :

(1) 

(E and Z)

(2) 

(3) 

(4) 

Q22

Reaction of R-2-butanol with p-toluenesulphonyl chloride in pyridine then LiBr gives:

(1) R-2-butyl bromide

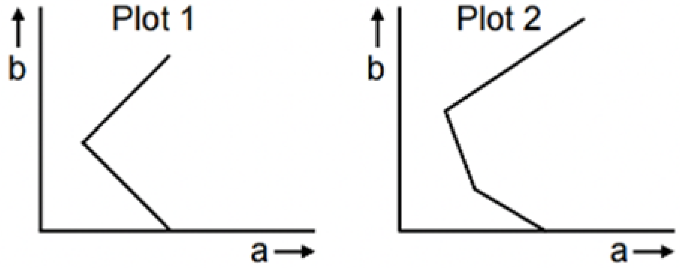
(2) S-2-butyl tosylate

(3) R-2-butyl tosylate

(4) S-2-butyl bromide

Q23

Given below are 2 conductometric titration plots. In each plot conductance of a solution in flask is plotted on x -axis (shown as a) and the volume of the titrant added from burette is plotted on y-axis (shown as b). Which of the following option(s) is/are correct?



(1) Plot 1 corresponds to titration of NaOH (in flask) versus HCl (in burette)

(2) Plot 2 corresponds to titration of NaOH (in flask) versus HCl (in burette)

(3) Plot 1 corresponds to titration of NaOH (in flask) versus mixed acid HCl + CH_3COOH (in burette)

(4) Plot 2 corresponds to titration of NaOH (in flask) versus mixed acid HCl + CH_3COOH (in burette)

Q24

The rate equation for the gaseous phase reaction $2AB + B_2 \rightarrow 2AB_2$, is observed to be Rate $= k[AB]^2\,[B_2]$ Possible mechanism(s) consistent with this rate equation is/are:

(1) $2\mathrm{AB} + \mathrm{B}_2 \xrightarrow{slow} 2\mathrm{AB}_2$

(2) $AB + AB \rightleftharpoons A_2B_2$ (fast equilibrium step), $A_2B_2 + B_2 \xrightarrow{slow} 2\mathrm{AB}_2$

(3) $AB + B_2 \xrightarrow{slow} A_3, \quad AB + AB_3 \xrightarrow{fast} 2A_2$

(4) $AB + B_2 \rightleftharpoons A_3$ (fast equilibrium step), $\quad AB + AB_3 \xrightarrow{slow} 2AB_2$

Q25

The commonly used drug paracetamol is synthesized using the following sequence:

1. Nitration of phenol with dilute HNO_3, producing mixture of X and Y.

2. X is removed from the product mixture by steam-distillation.

3. Reduction of Y by Sn/HCl to give Z.

4. Acetylation of Z using 1 equivalent of acetic anhydride in presence of pyridine to give paracetamol.

Which of the following is/are correct?

(1) Paracetamol is antipyretic and analgesic (pain-killer).

(2) X is ortho-nitrophenol

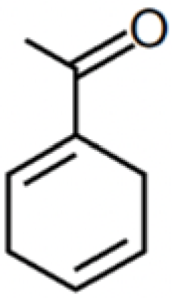
(3) Z is ortho-amino phenol

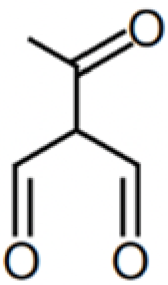
(4) During mono-acetylation of Z, $-NH_2$ group is acetylated in preference to $-OH$ group.

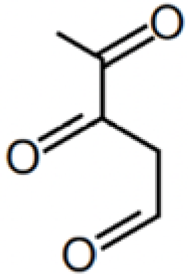
Q26

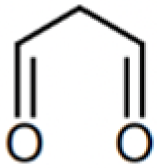
Which of the following option(s) is/are correct for the products of the given reaction sequence?

Acetophenone $\xrightarrow{\text{Na in liquid } NH_3} H \xrightarrow{O_3} I \xrightarrow{Zn/H_2O} J$ (contains 5C atoms) $+K$

(1) Compound H is 

(2) Compound J is 

(3) Compound J is 

(4) Compound K is 

Q27

Which of the following mixtures cannot be separated by passing H_2 S through their solutions containing dilute HCl ?

(1) Cu^{2+} and Sb^{3+}

(2) Pb^{2+} and Cd^{2+}

(3) Pb^{2+} and Al^{3+}

(4) Zn^{2+} and Mn^{2+}

Q28

Two litre of 1 molar solution of a complex salt $CrCl_3 \cdot 6H_2O$ (mol. wt. $= 266.5$) shows an osmotic pressure of 98.52 atm at 300 K. The solution is now treated with 1 litre of $6 \text{ M } AgNO_3$.
Which of the following is/are correct ? (Assume 100\% dissociation in complex compound,
$R = 0.082 \text{ atm. L/mol. K}$)

(1) Weight of AgCl precipitated is 861 gram.

(2) The clear solution obtained after treating with $AgNO_3$ will show an osmotic pressure of 98.5 atm.

(3) The clear solution obtained after treating with $AgNO_3$ will show an osmotic pressure of 65.6 atm.

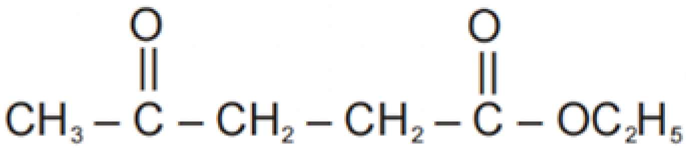
(4) 2 mol of $\left[Cr(H_2O)_6\right](NO_3)_3$ will be present in solution.

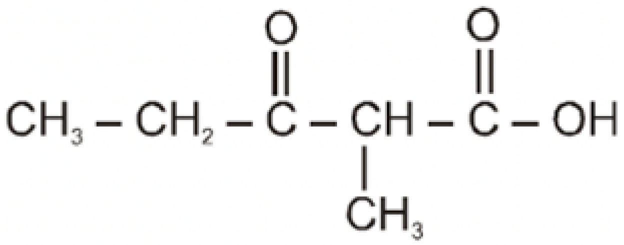
Q29

Which of the following will give 3-pentanone.

(1) $(\mathrm{CH_3-CH_2COO})_2\mathrm{Ca} \xrightarrow{\Delta}$

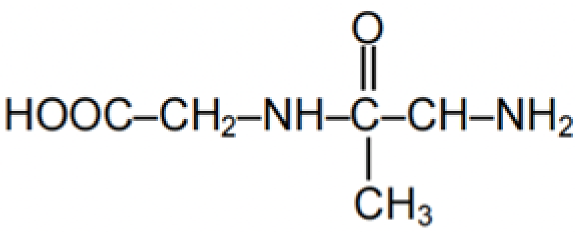
(2) $CH_3-CH_2-C\equiv N \xrightarrow[(2)\ H_3^{\oplus}O]{(1)\ CH_3-CH_2-MgBr}$

(3)  $\xrightarrow[(2)\ NaOH(CaO)]{(1)\ H_2O/H^+}$

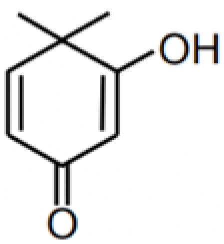
(4)  $\xrightarrow{\Delta}$

Q30

Select the correct statement(s).

(1) 

is a dipeptide of Glycine \& Alanine, whose abbreviated name is GLY-ALA

(2) Compound  show Keto-enol tautomerism.

(3) Phenol and benzoic acid can be distinguished by $\mathrm{NaHCO_3}$.

(4) Order of basicity in aqueous medium $\mathrm{MeNH_2} < \mathrm{Me_2NH} < \mathrm{Me_3\,N}$

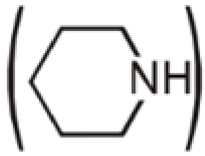
Q31

At 298 K and 1 bar partial pressure of hydrogen gas, the half cell reduction potential of hydrogen electrode is equal to -0.30 V at pOH $=?$(Take $2.303 \mathrm{RT/F} = 0.06$ V at 298 K)

Q32

(I) $In[\mathrm{Fe(edta)}]^{-}$, number of 5 membered rings is X.
(II) Let the total number of isomers exhibited by the complex ion $\left[\mathrm{Co(en)}_2\left(\mathrm{NO}_2\right)(\mathrm{CN})\right]^{+}$(including given isomer) is Y.
(III) Number of ambidentate ligand amongst the following is
$\left(\mathrm{I}^{-}, \mathrm{NO}_2^{-}, \mathrm{CN}^{-}, \mathrm{SCN}^{-}, \mathrm{C}_2\mathrm{O}_4^{2-}, \mathrm{NH}_3, \text{ en}, \mathrm{H}_2\mathrm{O}\right)$ Z.
Then determine value of $\frac{X+Y+Z}{4}$.

Q33

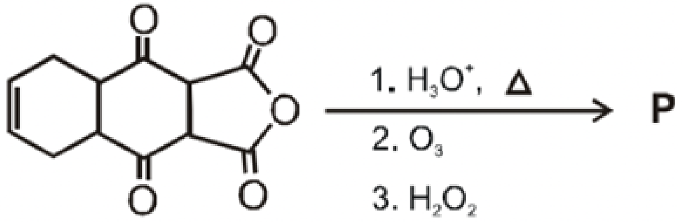
An organic compound (A) consists of C, H and O gives a characteristic colour with ceric ammonium nitrate. Treatment of (A) with PCl_5 gives (B) which reacts with KCN to from (C). The reduction of (C) with $\mathrm{Na/C_2H_5OH}$ produces (D) which on heating gives piperidine 
with the evolution of ammonia. Identify the number of carbon atoms in compound D is x and the number of carbon atoms in compound A is y, what is the value of $x - y$?

Q34

If the slope of Z (compressibility factor) v/s P curve is $\frac{\pi}{492.6}$ atm^{-1} at 300 K for a gas which shows negligible intermolecular force then the diameter of the gas molecule is _______ Å $\left(N_{\mathrm{A}} = 6 \times 10^{23},\right.$
$\left.\mathrm{R} = \frac{1}{12} \text{ atm L}^{-1} \text{ mol}^{-1} \text{ K}^{-1}\right)$

Q35

The total number of carboxylic acid groups in the product $\mathbf{P}$ is :



Q36

Diethyl β, β'-dimethyl glutarate (ester) is condensed with diethyl oxalate in presence of sodium ethoxide and ethanol to form a major product (P), which on acidic hydrolysis followed by heating gives another product (Q). Calculate total number of $\mathrm{C}=\mathrm{O}$ bonds in (P) and (Q) compounds?

Q37

If $\prod_{i=1}^{n} a_i = a_1 \cdot a_2 \cdot a_3 \ldots a_n$, then $\lim_{x \to 0} \frac{\left(\prod_{i=1}^{16} (1+ix)^{\frac{1}{2000+i}}\right)-1}{x}$ is equal to

(1) $\prod_{i=1}^{16} \frac{1}{2000+i}$

(2) $\sum_{\mathrm{i}=1}^{16} \frac{\mathrm{i}}{2000+\mathrm{i}}$

(3) $\sum_{i=1}^{16} \frac{i}{2000+i^2}$

(4) 1

Q38

For acute angle triangle ABC, the value of $\frac{\sin A}{A} + \frac{\sin B}{B} + \frac{\sin C}{C}$ is

(1) greater than $\frac{6}{\pi}$

(2) $\frac{6}{\pi}$

(3) less than $\frac{6}{\pi}$ but greater than $\frac{3}{\pi}$

(4) greater than 3

Q39

If $A(-1,2,-3), B(5,0,-6)$ and $C(0,4,-1)$ are the vertices of ΔABC, then direction ratios of the external bisector of $\angle BAC$ are

(1) -11, 20, 23

(2) $-11, 20, 20$

(3) 11,20,21

(4) none of these

Q40

ABCD is a cyclic quadrilateral with $\mathrm{AC} \perp \mathrm{BD}$ and O is the centre of its circumcircle, then $\overrightarrow{\mathrm{OA}} \cdot \overrightarrow{\mathrm{OB}} + \overrightarrow{\mathrm{OB}} \cdot \overrightarrow{\mathrm{OC}} + \overrightarrow{\mathrm{OC}} \cdot \overrightarrow{\mathrm{OD}} + \overrightarrow{\mathrm{OD}} \cdot \overrightarrow{\mathrm{OA}}$ is equal to

(1) 1

(2) -1

(3) 0

(4) none of these

Q41

With all angles measured in degree the product $\prod_{k=1}^{45} \operatorname{cosec}^2(2k-1) = m^n$ where $m,\ n$ are integers greater than 1 then

(1) $m+n=100$

(2) $\int_{-1}^{m} e^{\{mx\}} dx = 3\,(e-1)$ (where $\{.\}$ denotes fractional part function)

(3) Number of ways in which $n+5m+1$ can be resolved as product of two factors is 10

(4) Number of ways in which $m+n$ can be resolved as a product of two factors which are relatively prime is 2.

Q42

Let $A,\ B,\ C$ are three non-collinear points corresponding to complex numbers $Z_0 = ai,\ Z_1 = \frac{1}{2} + bi,\ Z_2 = 1 + ci$ respectively where $a, b, c \in R$. If L be a line bisecting AB and parallel to AC in ΔABC and a curve $C : Z = Z_0 \cos^4 t + 2Z_1 \cos^2 t \sin^2 t + Z_2 \sin^4 t$, where $t \in R$ then

(1) line L touches curve C

(2) Line L intersect curve C in real and distinct points

(3) Equation of line L is $y = (c - a)x + \frac{1}{4}(3a + 2b - c)$

(4) Line L neither touches nor intersect the curve C.

Q43

Let f : $\mathrm{R}^+ \rightarrow \mathrm{R}$ be a differentiable function with $\mathrm{f}(1) = 3$ and satisfying

$\int_1^{\mathrm{xy}} \mathrm{f}(\mathrm{t})\mathrm{dt} = \mathrm{y} \int_1^{\mathrm{x}} \mathrm{f}(\mathrm{t})\mathrm{dt} + \mathrm{x} \int_1^{\mathrm{y}} \mathrm{f}(\mathrm{t})\mathrm{dt}$ for all $x, y \in R^+$, then

(1) $\int_1^e f(x)dx = 3e$

(2) $\int_1^e f(x)dx = (e - 1)$

(3) $\int_1^e f'(x)dx = 3$

(4) $\int_1^e f'(x)dx = 3e$

Q44

Tangent at a point P_1 (other than $(0, 0)$) on the curve $y = x^3$ meets the curve again at P_2. The tangent at P_2 meets the curve again at P_3 and so on then

(1) Abscissa of $P_1, P_2, P_3, \ldots\ldots\ldots P_n$ form a G.P.

(2) Ordinates of $P_1, P_2, P_3, \ldots\ldots\ldots P_n$ from a G.P.

(3) $\operatorname{Lim}_{n \rightarrow \infty} \sum_{r=1}^{n} \frac{1}{x_r} = \frac{2}{3}$ where x_i is abscissa of P_i for $i = 1, 2, 3, \ldots\ldots\ldots\ldots$ with $x_1 = 1$

(4) $\frac{\text{Area of } \Delta P_2 P_3 P_4}{\text{Area of } \Delta P_1 P_2 P_3} = 16$

Q45

If the area bounded by the curve $y = \left|\cos^{-1}(\sin x)\right| - \left|\sin^{-1}(\cos x)\right|$ with x - axis from $\frac{3\pi}{2} \leq x \leq 2\pi$ is equal to $\frac{\pi^2}{k}, k \in N$ Then k is a multiple of

(1) 1

(2) 2

(3) 3

(4) 4

Q46

In a ΔABC, if $r=2, R=5$ and $r_3=12$ and Δ denotes the area of ΔABC. Also m is minimum value of the function $y=(x-2)(x-4)(x-6)(x-8)+16, x \in R$, Then $\Delta+m$ is divisible by

(1) 4

(2) 6

(3) 8

(4) 12

Q47

Which of the following is/are correct?

(1) The polynomial $(\cos \theta+x \sin \theta)^n-\cos n \theta-x \sin n \theta$ is divisible by $x^2+1, n \in N$.

(2) The polynomial $x^n \sin \theta-k^{n-1} x \sin n \theta+k^n \sin (n-1) \theta$ is divisible by $x^2-2 k x \cos \theta+k^2 \cdot k \in R-\{0\}$

(3) If $A\left(z_1\right), B\left(z_2\right), C\left(z_3\right)$ be vertices of an acute angle Δ in an argand plane having origin as its orthocentre, Then $z_1 \bar{z}_2+\bar{z}_1 z_2=z_2 \bar{z}_3+\bar{z}_2 z_3=z_3 \bar{z}_1+\bar{z}_3 z_1$

(4) $a^3+b^3+c^3-3 a b c=(a+b+c)\left(a \omega^3+b \omega^4+c \omega^5\right)\left(a \omega^6+b \omega^8+c \omega^7\right)$ Where ω is a complex cube root of unity.

Q48

A bag contains 10 balls of which some are black and others are white. A person draws 6 balls \& finds that 3 of them are white & 3 of them are black. Find probability that number of white balls in bag are same as black balls is $\frac{m}{n}$, (where m and n are relatively prime), then which of the following is/are True?

(1) $m+n=43$

(2) $m-n=23$

(3) $mn = 330$

(4) $m + n = 40$

Q49

If roots of $x^4 - 12x^3 + bx^2 + cx + 81 = 0$ are positive and $g(x)$ is inverse of $f(x) = x^{\alpha+2} + (\alpha+2)x + \alpha$ where α is root of $2bx + c = 0$, then find $6.\, g'(5)$

Q50

Let $f(x) = \left(x^2 + 3x + 2\right)^{\cos \pi x}$ find number of positive integers n for which $\left|\sum_{k=1}^{n} \log_{10} f(k)\right| = 1$

Q51

The possible number of ordered triplets (m, n, p) where $m, n, p \in N$ is $(6250k)$ such that $1 \leq \mathrm{m} \leq 100; 1 \leq \mathrm{n} \leq 50; 1 \leq \mathrm{p} \leq 25$ and $2^{\mathrm{m}} + 2^{\mathrm{n}} + 2^{p}$ is divisible by 3 then k is

Q52

The plane $6x + 3y + 2z = 6$ intersects x-axis, y-axis and z-axis at P, Q, R respectively. If the distance between origin and orthocenter of $\triangle ABC$ is equal to d, then 7 d is equal to

Q53

If $\left(1 + x + x^2\right)^{3n+1} = a_0 + a_1 x + a_2 x^2 + \ldots\, a_{6n+2} x^{6n+2}$, then find the value of $\sum_{r=0}^{2n} \left(a_{3r} - \frac{a_{3r+1} + a_{3r+2}}{2}\right)$ is

Q54

In ΔABC, if $\frac{c+a}{12} = \frac{a+b}{14} = \frac{b+c}{18}$, then find the value of $\frac{7r_1 - r_2}{11r}$ is.

Answer Key

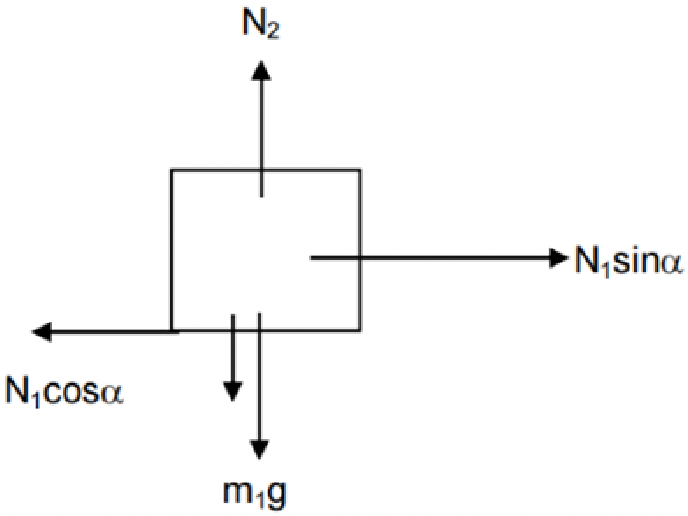
Q1 (3)	**Q2** (2)	**Q3** (1)	**Q4** (1)
Q5 (2) (4)	**Q6** (1) (2)	**Q7** (1) (2) (4)	**Q8** (1) (2) (3)
Q9 (1) (3)	**Q10** (1) (2) (3) (4)	**Q11** (2) (3)	**Q12** (1) (4)
Q13 (5)	**Q14** (6)	**Q15** (6)	**Q16** (3)
Q17 (4)	**Q18** (5)	**Q19** (2)	**Q20** (1)
Q21 (2)	**Q22** (4)	**Q23** (1) (3)	**Q24** (1) (2) (4)
Q25 (1) (2) (4)	**Q26** (2) (4)	**Q27** (1) (2) (4)	**Q28** (1) (3) (4)
Q29 (1) (2) (4)	**Q30** (2) (3)	**Q31** (9)	**Q32** (5)
Q33 (2)	**Q34** (5)	**Q35** (2)	**Q36** (6)
Q37 (2)	**Q38** (1)	**Q39** (1)	**Q40** (3)
Q41 (2) (4)	**Q42** (1) (3)	**Q43** (1) (3)	**Q44** (1) (2) (3) (4)
Q45 (1) (2) (4)	**Q46** (1) (2) (3) (4)	**Q47** (1) (2) (3) (4)	**Q48** (1) (3)
Q49 (1)	**Q50** (2)	**Q51** (5)	**Q52** (6)
Q53 (0)	**Q54** (0)		

Q1

For rotational equilibrium of rod about A

$Mg\cos\alpha S+m_2g\frac{\ell}{2}\cos\alpha=N_1\ell$

For Translational equilibrium of block



$N_2=m_1g+N_1\cos\alpha$

$N_1\sin\alpha=\mu N_2$

Solving above equation we get

$S=\left(\frac{\mu m_1}{\sin\alpha-\mu\cos\alpha}-\frac{m_2\cos\alpha}{2}\right)\frac{\ell}{M\cos\alpha}.$

Q2

$\lambda_{\text{th}}=\frac{hc}{eV_a}$

$\frac{1}{\lambda_{K\alpha}}=R(z-1)^2\left(\frac{1}{1^2}-\frac{1}{2^2}\right)$

$\frac{13}{10}(\lambda_{K_\alpha}-\lambda_{\text{th}})=\left(\lambda_{K_\alpha}-\frac{\lambda_{\text{th}}}{2}\right)$

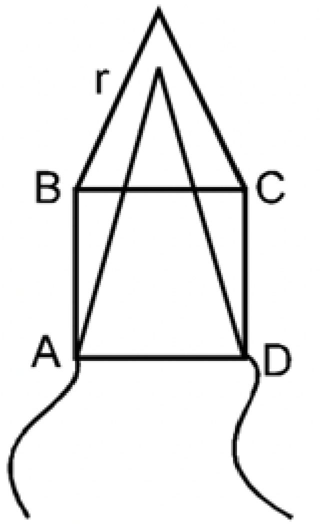
$\frac{3}{10}\lambda_{K_\alpha}=\left(\frac{13}{10}-\frac{1}{2}\right)\lambda_{\text{th}}$

$\frac{3}{10}\left(\frac{4\times10^{-7}}{3(Z_7)^2}\right)=\left(\frac{8}{10}\right)\frac{12.4\times10^{-7}}{15.5\times10^3}\Rightarrow\frac{5000}{8}$

$=(z-1)^2$

$625=(z-1)^2\Rightarrow z=26$

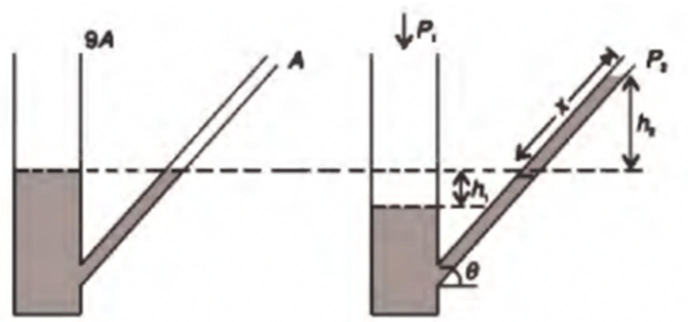
Q3



$R_{eq} = \frac{\frac{8r}{3} \times \frac{2r}{3}}{\frac{10r}{3}} = \frac{8r}{15} = 8\Omega$

Q4

When a pressure difference $P_1 - P_2 = \Delta P$ is applied various height and length are as shown in the second figure.



Since volume of manometer liquid is constant, we have

$\therefore 9Ah_1 = A_x \Rightarrow x = 9h_1$

$h_2 = x \sin\theta = 9h_1 \sin\theta$

$\Delta P = \rho g (h_1 + h_2)$ [$\rho =$density of liquid]

$= 0.74\rho_w g \left[\frac{x}{9} + x\sin\theta\right] = 0.74\rho_\omega g x \left[\frac{1}{9} + \sin\theta\right]$

It is needed that when $\Delta P = 0.09 \times 10^{-3}\rho\omega g$, value of x is $0.5 \times 10^{-3}m$

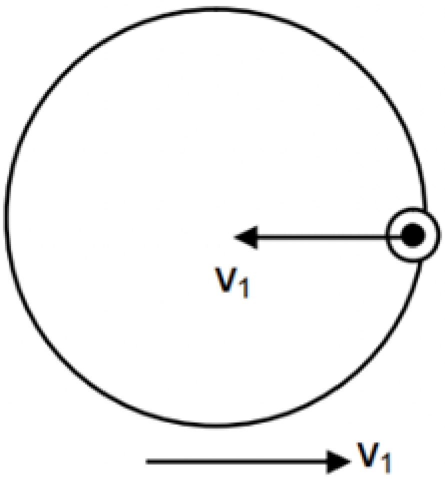
$\therefore 9 \times 10^{-5} \times \rho_\omega g = 0.74\rho_\omega g \times 5 \times 10^{-1}\left[\frac{1}{9} + \sin\theta\right]$

$\frac{0.9}{0.74 \times 5} = \frac{1}{9} + \sin\theta$

$\sin\theta = 0.24 - 0.11 = 0.13$

$\therefore\ \theta = \sin^{-1}(0.13)$

Q5



(A) $\mathrm{m}(3\mathrm{R} - \mathrm{x}) - \mathrm{mx} = 0 \Rightarrow \mathrm{x} = \frac{3\mathrm{R}}{2}$

(B) By conservation of energy

$2 \times \frac{1}{2}mv_1^2 - \frac{kQq}{R} = \frac{-kQq}{2R}$

$mv_1^2 = \frac{kQq}{2R} \Rightarrow v_1 = \sqrt{\frac{kQq}{2mR}}$

(C)$t_{AB} = \frac{2R}{2\sqrt{\frac{kQq}{2mR}}} = \sqrt{\frac{2mR^3}{kQq}}$

(D)Total distance covered by shell and particle with constant velocity is 2R, therefore distance covered by shell with constant velocity is R.

Q6

The decay scheme is as shown

$A \xrightarrow{\lambda_1=\lambda} B \xrightarrow{\lambda_2=\lambda+2\lambda} C + D\,(\text{stable})$

$t = 0 \quad N_0 \quad 0 \quad 0$

$t \quad N_1 \quad N_2 \quad N_3 + N_4$

Here $\mathrm{N}_1,\ \mathrm{N}_2$ and $+\mathrm{D}$ at any time t.

For A, we can write

$N_1 = N_0 e^{-\lambda\,1t}$ ... (1)

For B, we can write

$\frac{dN_2}{dt} = \lambda_1 N_1 - \lambda_2 N_2$ ... (2)

or, $\frac{dN_2}{dt} + \lambda_2 N_2 = \lambda_1 N_1$

This is a linear differential equation with integrating factor

I.F. $= e^{\lambda_2 t}$

$e^{\lambda_2 t}\frac{dN_2}{dt} + e^{\lambda_2 t}\lambda_2 N_2 = \lambda_1 N_1 e^{\lambda_2 t}$

$\int d\left(N_2 e^{\lambda_2 t}\right) = \int \lambda_1 N_1 e^{\lambda_2 t} dt$

$N_2 e^{\lambda_2 t} = \lambda_1 N_0 e^{\lambda_2 t}$ ... using (1)

$N_2 e^{\lambda_2 t} = \lambda_1 N_0 \frac{e^{(\lambda_2-\lambda_1)t}}{\lambda_2-\lambda_1} + C$ ... (3)

At$t = 0,\ N_2 = 0\ 0 = \frac{\lambda_1 N_0}{\lambda_2-\lambda_1} + C$

Hence $C = \frac{\lambda_1 N_0}{\lambda_1-\lambda_2}$

Using C in eqn. (3), we get

$N_2 = \frac{\lambda_1 N_0}{\lambda_2-\lambda_1}\left(e^{-\lambda_1 t} - e^{-\lambda_2 t}\right)$

and$N_1 + N_2 + N_3 + N_4 = N_0$

$\therefore N_3 + N_4 = N_0 - (N_1 + N_2)$

Q7

$\mathrm{T} = \frac{\mathrm{mv}^2}{\mathrm{R}} = \frac{2\mathrm{E}_0}{\mathrm{R}}$

$\mathrm{L} = \mathrm{mvR}$

so $\mathrm{T} = \frac{\mathrm{L}^2}{\mathrm{mR}^3}$

when radius of circle is r then $\mathrm{T}' = \frac{\mathrm{L}^2}{\mathrm{mr}^3}$

work done $= -\int_R^{R/2} T'\,dr = \frac{3\,L^2}{2mR^3} = 3E_0$

Q8

Pressure in left bubble $= \frac{4s}{r_1}$

Volume of left bubble $= \frac{4}{3}\pi r_1^3$

Pressure in right bubble $= \frac{4\,\mathrm{S}}{r} = \frac{4}{3}\pi r_2^3$

$\mathrm{n}_1 + \mathrm{n}_2 = 2\mathrm{n}$

$\mathrm{r}_1{}^2 + \mathrm{r}_2^2 = 2\mathrm{R}^2$

$r_1{}^2 + r_2^2 = 2R^2$

$\mathrm{A}_1 + \mathrm{A}_2$ and $\mathrm{n}_1 + \mathrm{n}_2$ are constant

$\frac{\mathrm{n}_1}{\mathrm{n}_2} = \frac{\mathrm{r}_1^2}{\mathrm{r}_2^2} = \frac{4}{1} \Rightarrow \mathrm{r}_2^2 = \frac{2\mathrm{R}^2}{5}$

$r_2 = \sqrt{\frac{2}{5}}R$

$r_1 = \sqrt{\frac{8}{5}}R$

Q9

$$(2\mathrm{n}-1)\frac{\mathrm{v}}{4\ell} = 264$$

$$\Rightarrow \ell = \frac{(2\mathrm{n}-1)\times 330}{264\times 4}$$

$$\Rightarrow \ell = 31.25 \text{ if } \mathrm{n} = 1$$

$\Rightarrow \ell = \frac{(2\mathrm{n}-1)\times 330}{264\times 4}$

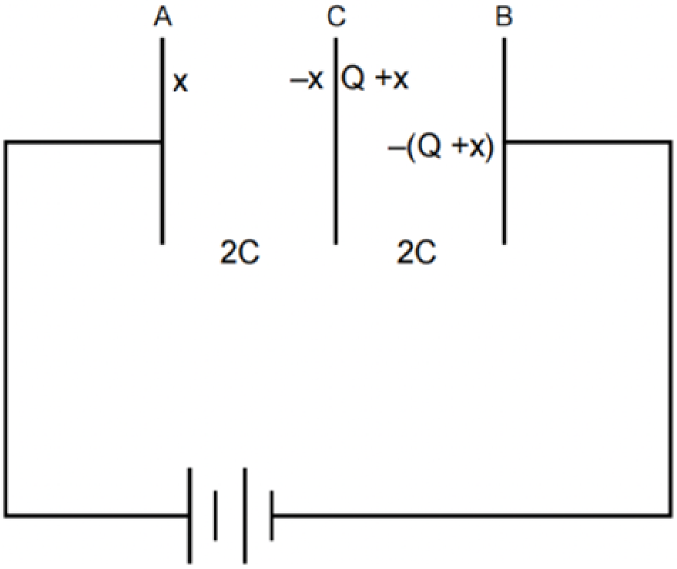
$\Rightarrow \ell = 31.25$ if $\mathrm{n} = 1$

$\ell = 93.75$ if $\mathrm{n} = 2$

$\ell = 156.25$ if $\mathrm{n} = 3$

$\ell = 156.25$ if $\mathrm{n} = 3$

Q10



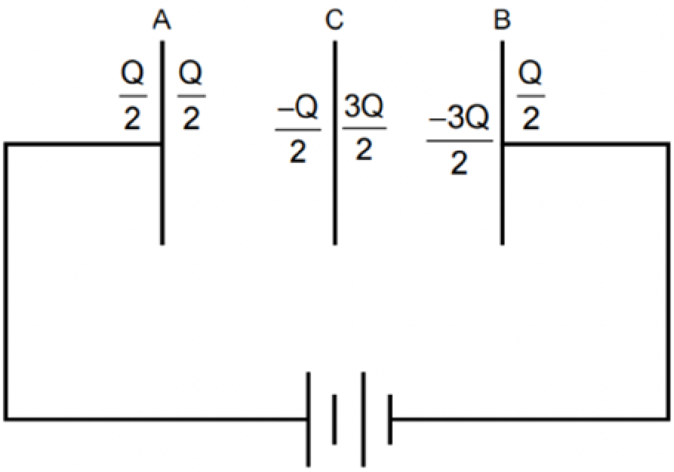
C = capacitance of plates A & B combined

$\Rightarrow \frac{x}{2C}+\frac{Q+x}{2C}=V$,[KVL]

$\Rightarrow Q+2x=2CV$,

$[CV=Q$, initially]

$\Rightarrow Q+2x=2Q$



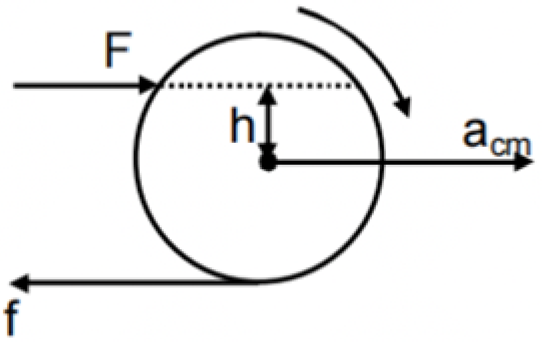
$\Rightarrow x=\frac{Q}{2}$

$\rightarrow$ Outer plates of A &B will have $\frac{Q}{2}$ charge each. [no net charge in the circuit]

Q11

$F - f = ma_{cm}$ ... (1)

$Fh + fR = \frac{2}{5}mR^2\frac{a_{cm}}{R}$ ... (2)



(1)& (2) $a_{cm} = \frac{5F}{7M}\left[1 + \frac{h}{R}\right]$

$(a_{cm})_{\max} = \frac{10\ F}{7M} \quad f_{\max} = \frac{3\ F}{7}$

If f = 0, F = ma

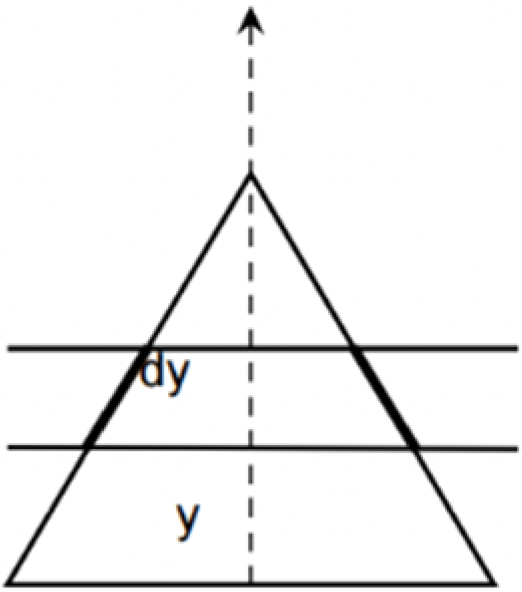
Fh = 2/5 x maR

h = 2R/5

Q12

dF on each of the two symmetric elements

$= I \times \left(\frac{2dy}{\sqrt{3}}\right) B_0 y \times \frac{\sqrt{3}}{2}$



$$\Rightarrow \tau = \int_0^{\frac{\sqrt{3}a}{2}} \left(\frac{2B_0 I_y dy}{\sqrt{3}}\right) \times \left(\frac{\sqrt{3}}{2}a - y\right)$$

$$= \frac{2}{\sqrt{3}} I B_0 \int^{\frac{\sqrt{3}}{2}a} \left(\frac{\sqrt{3}}{2}a - y\right) y\, dy = \frac{I B_0 a^3}{8}$$

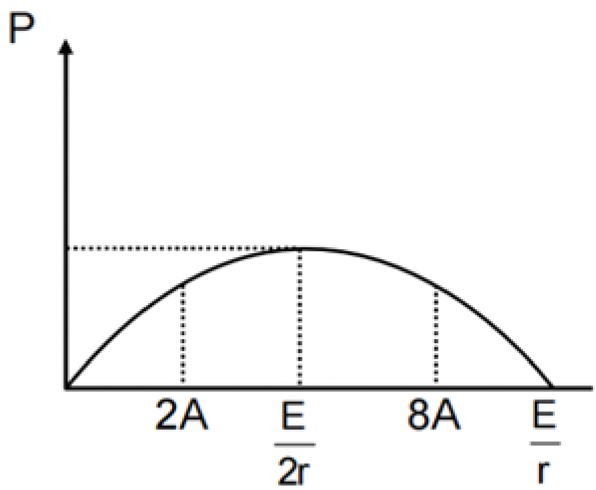
$$\tau_{\text{friction}} = 2\left\{\frac{\mu m g}{a}\int_0^{a/2} x\, dx\right\} = \frac{\mu\, m g a}{4}$$

For the frame to rotate

$$\frac{I B_0 a^3}{8} \le \frac{\mu m g a}{4}$$

$$\Rightarrow B_0 \le \frac{2\mu m g}{I a^2}$$

Q13



$P = Ei - i^2 r$

$i^2 r - Er + P = 0$

Sum of roots.

$i_1 + i_2 = \frac{E}{r} = 10$ A

For maximum power $i = \frac{E}{2r} = 5A$

Q14

$\mathrm{N_1} = 2\ \mathrm{N_2}$

$\mathrm{N_0 e^{-\lambda_1 t}} = 2\ \mathrm{N_0 e^{-\lambda_2 t}}$

$\Rightarrow \dfrac{1}{\mathrm{t}} = \dfrac{1}{\mathrm{T_2}} - \dfrac{1}{\mathrm{T_1}}$

$N_0 e^{-\lambda_1 t} = 2\ N_0 e^{-\lambda_2 t}$

$\Rightarrow \frac{1}{t} = \frac{1}{T_2} - \frac{1}{T_1}$

t = 6

Q15

Torque due to magnetic force should act opposite to that of gravity i.e. along the-ve y-axis. If $\mathbf{M}\hat{k}$ is the magnetic moment

$\vec{\tau}_{\mathrm{B}} = \overrightarrow{\mathrm{M}} \times \overrightarrow{\mathrm{B}} = \mathrm{M}\hat{\mathrm{k}} \times (3\hat{\mathrm{i}} + 4\hat{\mathrm{k}})\mathrm{B_0} = 3\mathrm{MB_0}\hat{\mathrm{j}}$

$\Rightarrow M$ is $-ve$

$\vec{\tau}_{\mathrm{B}} = \overrightarrow{\mathrm{M}} \times \overrightarrow{\mathrm{B}} = \mathrm{M}\hat{\mathrm{k}} \times (3\hat{\mathrm{i}} + 4\hat{\mathrm{k}})\mathrm{B_0} = 3\mathrm{MB_0}\hat{\mathrm{j}}$

$\therefore$ I should be clockwise i.e. from P to Q

$\overrightarrow{F} = I(\overrightarrow{L} \times \overrightarrow{B}) \Rightarrow$

$\overline{F}_{RS} = I\left[(-b\hat{j}) \times (3\hat{i} + 4\hat{k})B_0\right] = I\ B_0\ b[3\hat{k} - 4\hat{i})$

$3(abl)B_0 = mga/2$

$I = \frac{mg}{6\ B_0\ b}$

Q16

Let the current be in the outer coil

The field at centre $B = \frac{\mu_0 l}{2b}$

The flux through the inner coil $= \frac{\mu_0 |\pi a^2}{2b}$

The induced emf produced in the inner coil

$\varepsilon = -\dfrac{d\phi}{dt}$

$\dfrac{\mu_0 \pi a^2}{2b} \dfrac{d}{dt}\left(2t^2\right) = \dfrac{2\mu_0 \pi a^2 t}{b}$

$\frac{\mu_0 \pi a^2}{2b} \frac{d}{dt}\left(2t^2\right) = \frac{2\mu_0 \pi a^2 t}{b}$

Current induced in the inner coil $= \frac{\varepsilon}{\mathrm{R}} = \frac{2\mu_0 \pi \mathrm{a^2 t}}{\mathrm{bR}}$

Heat developed in the inner coil $= \int_0^t I^2 R dt$

$= \int_0^t \frac{4\mu_0^2\pi^2 a^4 t^2 R dt}{b^2 R^2} = \frac{4\mu_0^2\pi^2 a^4}{b^2 R}\frac{t^3}{3}$

Q17

For the lens

$$\frac{1}{v} - \frac{1}{-36} = \frac{1}{30}$$

$\therefore \quad v = 180 \text{ cm}$

$\therefore \quad v = 180 \text{ cm}$

This acts as an (virtual) object for the mirror which forms its image at a distance of 80 cm from it (in the absence of water tray).

The water tray shifts the image by

$t\left(1 - \frac{1}{\mu}\right) = h\left(1 - \frac{3}{4}\right)$

$\therefore \quad \mathrm{h}/4 + 80 = 85 \text{ cm} \Rightarrow \mathrm{h} = 20 \text{ cm}$

$\therefore \quad h/4 + 80 = 85\ cm \Rightarrow h = 20\ cm$

Q18

Velocity of ball m, just after collision with ground $\mathrm{u}_1 = \sqrt{4\mathrm{gr}}$

Velocity of ball m_2 at this instant is $\mathrm{u}_2 = -\sqrt{4\mathrm{gr}}$

Velocity of m_2 just after collision between m_1 & m_2

$v_2 = -u_2 + 2u_1 = \sqrt{4gr} + 2\sqrt{4gr} = 3\sqrt{4gr}$

max height by $m_2 = 2r + \frac{v_2^2}{2g} = 20r$

Q19

$\frac{\mathrm{d_{bcc}}}{\mathrm{d_{ccp}}} = \frac{\text{packing efficiency of bcc}}{\text{packing efficiency of ccp}} = \frac{67.92}{74.02} = 0.918$

Q20

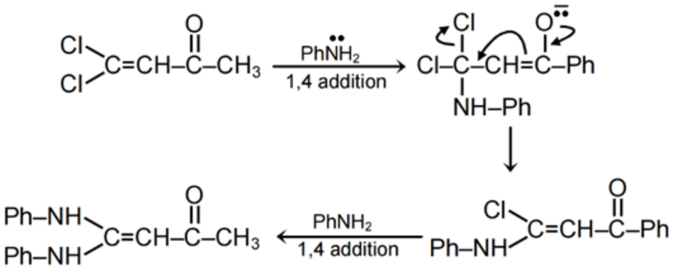
(A)Propene $\xrightarrow[\text{2. } H_2O_2/OH^-]{\text{1. } B_2H_6/THF}$ Propan-1-ol

(B)Propyne $\xrightarrow{HgSO_4/dil.H_2SO_4}$ Propan-2-ol

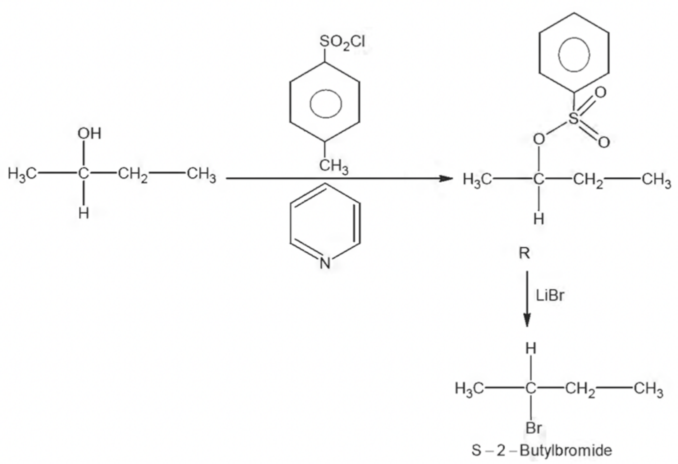
(C)Propene $\xrightarrow{dilute\ H_2SO_4}$ Propan-2-ol

(D)$CH_3COCl \xrightarrow{H_2/Pd/BaSO_4} CH_3CHO$

Q21



Q22



Q23

For titration of HCl v/s NaOH, conductance first decreases and then increases.

For titration of mixed acid $HCl + CH_3COOH$ v/s NaOH, also conductance first decreases and then increases.

Q24

In mechanism each step is elementary whose rate law can be written directly from stoichiometric coefficients of reactants. Rate of reaction = rate of slowest step.

Rate law should not contain concentration of intermediate. It may contain concentration of reactant, product and catalyst.

A.Rate $= k[\mathrm{AB}]^2\left[\mathrm{B}_2\right]$

B.Rate = rate of slowest step $= k\left[B_2\right]\left[A_2 B_2\right]$

$= k\left[B_2\right] K_{\mathrm{eq}}[AB][AB] = k'[AB]^2\left[B_2\right]$

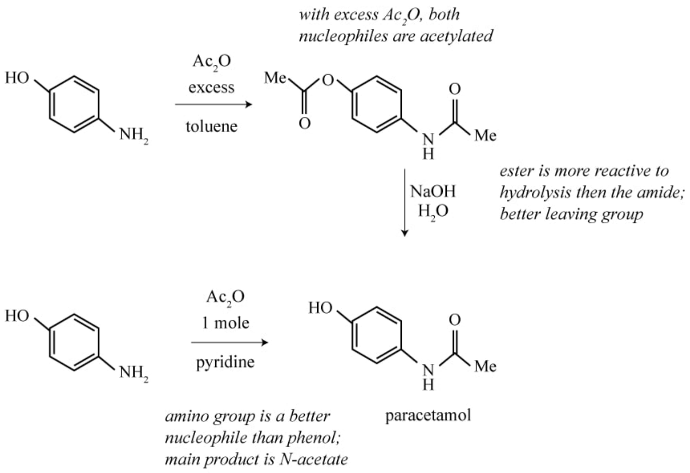
C.Rate $= k[\mathrm{AB}]\left[\mathrm{B}_2\right]$

D.Rate = rate of slowest step $= k\left[\mathrm{AB}^2\right]\left[\mathrm{AB}_3\right]$

$= k[AB]k_{eq}[AB][B_2] = k'[AB]^2[B_2]$

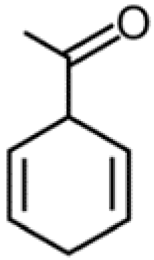
Q25

Nitration of phenol with dilute HNO_3, gives mixture of o-nitrophenol and p-nitrophenol; of which o-nitrophenol (X) is steam-volatile. Y on reduction gives p-amino phenol.



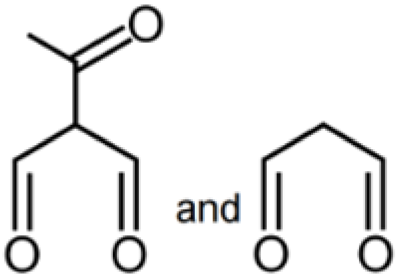
Paracetamol is antipyretic and analgesic (pain-killer).

Q26

Acetophenone $\xrightarrow[\text{Birch reduction}]{Na,\ Liq.NH_3}$ 
H

(obtained by 1,4 reduction at ipso-para position to keto group because keto group is electron withdrawing group)

H forms ozonide intermediate (I) with ozone, which undergoes reduction by Zn to form



Q27

(i)Cations of same group cannot be separated by the group reagent.

(ii)A cation is generally precipitated by the group reagent of a latter group. H_2 S in acidic medium is a reagent for group II.

(A)and (B) cannot be separated as they belong to group II.

Q28

$\mathrm{CrCl_3 \cdot 6H_2O}$

$\pi = \mathrm{CST}(1 - \alpha + x\alpha + y\alpha)$

$98.52 = 1 \times 0.082 \times 300 \times (x + y)[\because \alpha = 1]$

$x + y = 4$

$\left[\mathrm{Cr(H_2O)_6}\right]\mathrm{Cl_3} \rightleftharpoons \left[\mathrm{Cr(H_2O)_6}\right]^{3+} + 3\mathrm{Cl}^-$

$$\begin{array}{ccc} 1 & 0 & 0 \\ 1-\alpha & \alpha & 3\alpha \end{array}$$

ol of $AgNO_3$ will react with 1 mol of $\left[Cr(H_2O)_6\right]_3$

$\left[Cr(H_2O)_6\right]Cl_3 + 3AgNO_3 \rightarrow \left[Cr(H_2O)_6\right](NO_3)_3 + 3AgCl \downarrow$

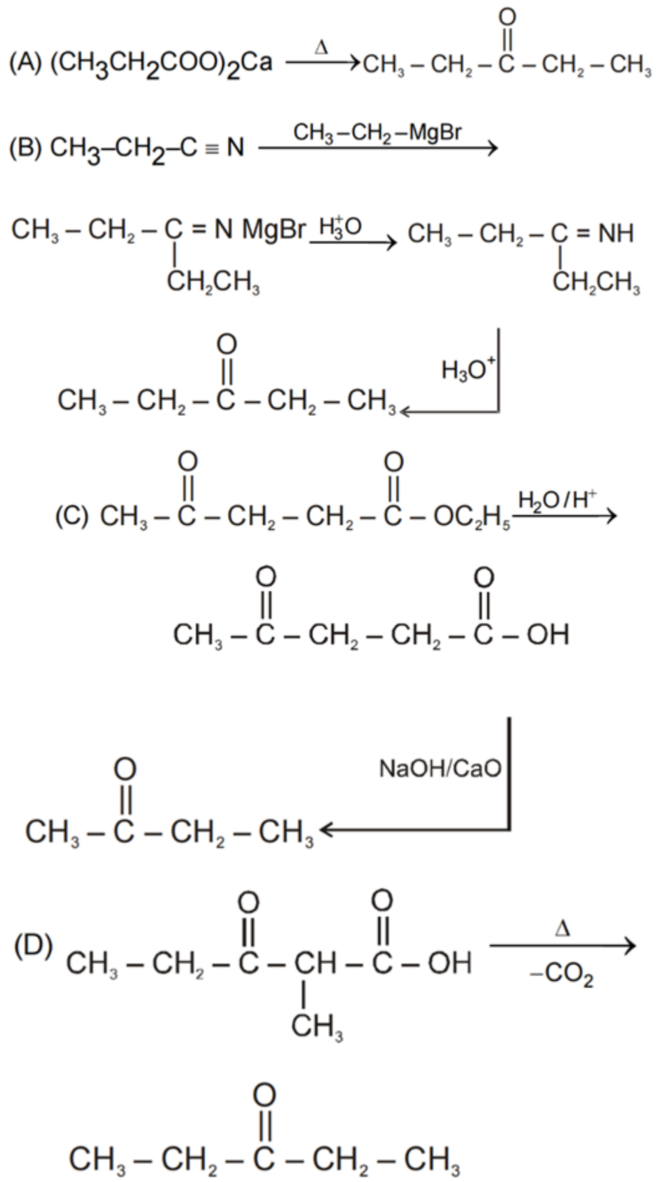
Moles of AgCl formed $= 6$

Wt. of AgCl formed $= 6 \times 143.5 = 861$ g

$\pi = \mathrm{CST} \times (1 + 3\alpha)$

$= 2/3 \times 0.082 \times 300 \times 4 = 65.6$ atm

Q29



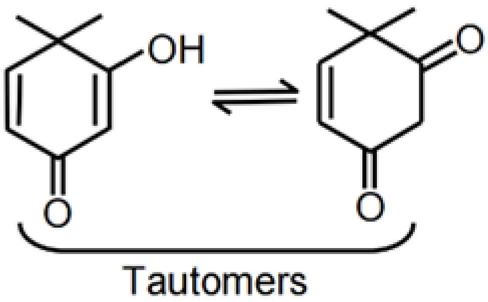
Q30

(A)Incorrect

In abbreviated form of dipeptides, amino acid having free amino group is written first so

Abb. From : ALA-GLY not GLY-ALA

(B)Correct



(C)Correct

$\mathrm{Ph-COOH+NaHCO_3 \rightleftharpoons Ph-COONa+H_2CO_3 \rightarrow H_2O+CO_2 \rightarrow Ph-OH+NaHCO_3 \rightarrow}$ No reaction

(D) Order of basicity in aqueous medium in methylamines is secondary>primary>tertiary. But in the case of ethylamines it is secondary>tertiary>primary.

Q31

Reduction half reaction for hydrogen electrode is

$\mathrm{H^+ + e^- \rightarrow 1/2H_2}$.

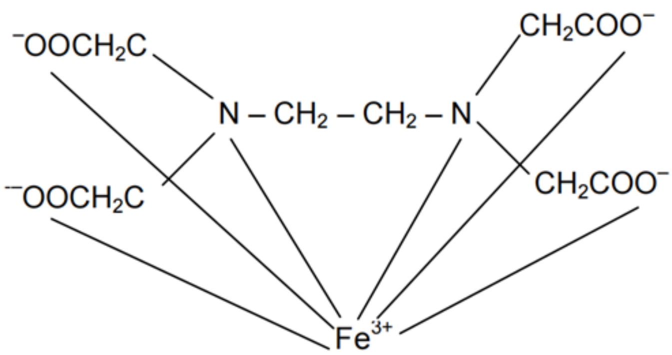
If p is partial pressure of hydrogen gas, reduction potential of hydrogen electrode as per Nernst equation is

$E = E^{\circ} - 0.06 \log\{p^{1/2}/[\mathrm{H^+}]\} = 0 - 0.06 \times \mathrm{pH} = -0.30\ \mathrm{V}$

Thus, $\mathrm{pH} = 5$ and $\mathrm{pOH} = 9$

Q32

(I) $X = 5$



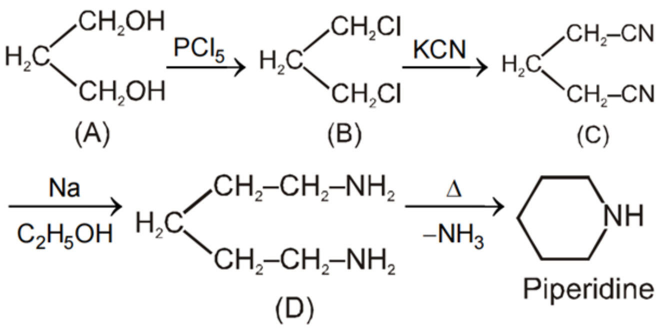
(II) $Y = 12$

(III) $Z = 3$ ambidentate - $\mathrm{NO}_2^-, \mathrm{CN}^-, \mathrm{SCN}^-$

Q33

(A)$\Rightarrow$ Colour with cerric ammonium nitrate

$\Rightarrow$ Alcoholic group.



Q34

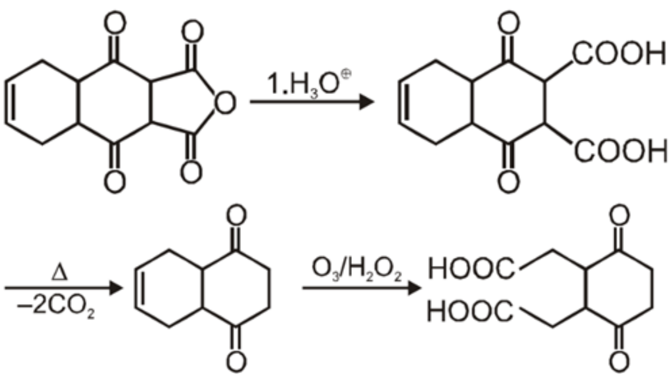
$\mathrm{Z} = 1 + \frac{\mathrm{Pb}}{\mathrm{RT}}$ at high pressure

$b = \frac{\pi}{492.6} \times \frac{1}{12} \times 300$

$b = \frac{4}{3}\pi r^3 \times 4N_A = \frac{\pi}{492.6} \times \frac{1}{12} \times 300\ \mathrm{L/mol}$

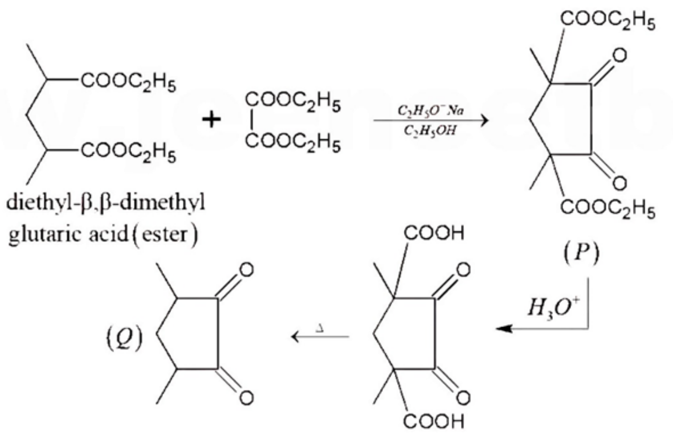
$r = 2.5\text{Å} \quad d = 5\text{Å}$

Q35



Number of $-COOH$ group is 2.

Q36



Q37

$\lim_{x \to 0} \frac{(1+x)^{\frac{1}{2001}}(1+2x)^{\frac{1}{2002}}\ldots\ldots(1+16x)^{\frac{1}{2016}}-1}{x}$

$= \lim_{x \to 0} \frac{f(x)-f(0)}{x-0} = f'(0)$

$f(x) = (1+x)^{\frac{1}{2001}}(1+2x)^{\frac{1}{2002}}\ldots\ldots(1+16x)^{\frac{1}{2016}}$

$\ell n f(x) = \sum_{i=1}^{16} \frac{1}{(2000+i)} \ln(1+ix)$

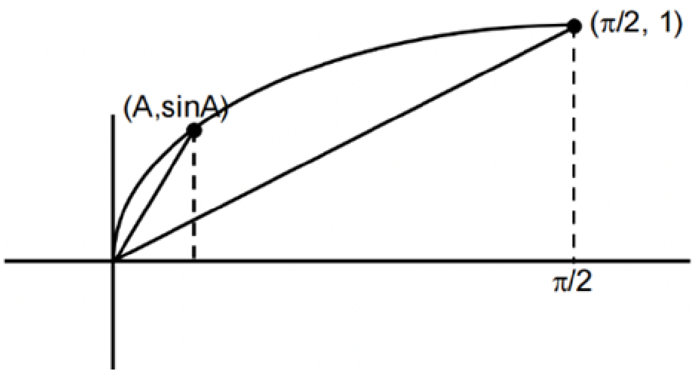
differentiate w.r.t x and put $x=0$

$\frac{1}{f(x)} f'(x) = \sum_{i=1}^{16} \frac{1}{(2000+i)} \frac{1}{(1+ix)} \cdot i$

$\Rightarrow f'(0) = \sum_{i=1}^{16} \frac{i}{2000+i}$

Q38

Clearly $\frac{\sin A}{A} > \frac{2}{\pi}$



$\Rightarrow \frac{\sin A}{A} + \frac{\sin B}{B} + \frac{\sin C}{C} > \frac{6}{\pi}$

Q39

If the external bisector of $\angle BAC$ meets BC at E, then E divides B C externally in the ratio of AB : AC.

$AB = 7, AC = 3$

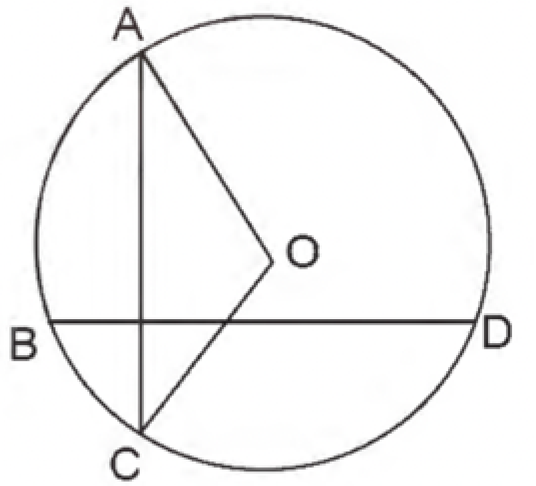
$\therefore$ Coordinates of E are $\left(\frac{-15}{4}, 7, \frac{11}{4}\right)$

$\therefore$ The direction ratio of line AE are $\frac{-15}{4} + 1, 7 - 2, \frac{11}{4} + 3 = -\frac{11}{4}, 5, \frac{23}{4}$

$= -11, 20, 23$

Q40

As $\mathrm{OA} = \mathrm{OC}$



so $\overrightarrow{\mathrm{OA}} + \overrightarrow{\mathrm{OC}}$ will be perpendicular to AC

$\therefore \overrightarrow{\mathrm{OA}} + \overrightarrow{\mathrm{OC}} = \lambda \overrightarrow{\mathrm{BD}}$ ... (1)

where λ is the real number.

$\overrightarrow{\mathrm{OB}} + \overrightarrow{\mathrm{OD}} = \mu \overrightarrow{\mathrm{AC}}$ ... (2)

where μ is the real number.

Now using (1) and (2), we get

$(\overrightarrow{\mathrm{OA}} + \overrightarrow{\mathrm{OC}})(\overrightarrow{\mathrm{OB}} + \overrightarrow{\mathrm{OD}}) = \mu . \lambda \overrightarrow{\mathrm{AC}} \cdot \overrightarrow{\mathrm{BD}} = 0$

Q41

$\prod_{k=1}^{45} \operatorname{cosec}^2(2k-1)$

$= \operatorname{cosec}^2 1^\circ \cdot \operatorname{cosec}^2 3^\circ \ldots\ldots \operatorname{cosec}^2 89^\circ$

$= \frac{1}{\sin^2 1^\circ \cdot \sin^2 3^\circ \ldots\ldots \sin^2 89^\circ}$

$= \frac{1}{(\sin 1^\circ \sin 3^\circ \sin 5^\circ \ldots\ldots\ldots \sin 89^\circ)(\sin 1^\circ \sin 3^\circ \ldots\ldots \sin 89^\circ)}$

$= \frac{1}{(\sin 1^\circ \sin 3^\circ \sin 5^\circ \ldots\ldots\ldots \sin 89^\circ)(\sin 179^\circ \sin 177^\circ \ldots\ldots \sin 91^\circ)}$

$= \frac{1}{\sin 1^\circ \sin 3^\circ \sin 5^\circ \ldots\ldots\ldots \sin 179^\circ}$

$= \frac{\sin 2^\circ \sin 4^\circ \sin 6^\circ \ldots\ldots\ldots \sin 178^\circ}{\sin 1^\circ \sin 2^\circ \sin 3^\circ \sin 4^\circ \ldots\ldots\ldots \sin 178^\circ \cdot \sin 179^\circ}$

$= \frac{2^{89} \cos 1 \cdot \cos 2 \cdot \cos 3 \ldots\ldots \cos 89}{\sin 91 \cdot \sin 92 \ldots\ldots\ldots \sin 179} = 2^{89}$

$\Rightarrow\ m = 2, \quad n = 89\ m + n = 91$

$\int_{-1}^{2} e^{\{2x\}} dx == \frac{1}{2} \int_{-2}^{4} e^{\{x\}} dx$

$= \frac{1}{2} \times 6 \times \int_0^1 e^{\{x\}} dx \quad 3(e-1)$

$n + 5m + 1 = 100 = 2^2 5^2 \Rightarrow$ Number of ways

$= \frac{1}{2}[(2+1)(2+1)+1] = 5; \quad m + n = 91 = 7 \times 13$

$\Rightarrow$ Required number of ways $= 2^{2-1} = 2$

Q42

$C : Z = Z_0 \cos^4 t + 2z_1 \cos^2 t \sin^2 t + z_2 \sin^4 t$

Let $z = x + iy$

$x + iy = (ai)\cos^4 t + 2\left(\frac{1}{2} + bi\right)\cos^2 t \cdot \sin^2 t + (1 + ci)\sin^4 t$

$\Rightarrow x = \cos^2 t \cdot \sin^2 t + \sin^4 t = \sin^2 t$ and $y = \mathrm{a}\cos^4 t + 2b\cos^2 t \sin^2 t + \mathrm{c}\sin^4 t$

$\Rightarrow \mathrm{y} = \mathrm{a}(1-\mathrm{x})^2 + 2\ \mathrm{b}(1-\mathrm{x})\mathrm{x} + \mathrm{cx}^2$

$\Rightarrow y = (a + c - 2b)x^2 + 2(b-a)x + a(0 \leq x \leq 1)$

$\because$ A, B, C are non-collinear

$\therefore a + c - 2b \neq 0$

$\therefore$ equation (i) is segment of parabola.

Now mid points of A B and B C are

D $\left(\frac{1}{4}, \frac{\mathrm{a+b}}{2}\right)$ and E $\left(\frac{3}{4}, \frac{\mathrm{b+C}}{2}\right)$ respectively.

$\therefore$ Equation of DE

$y=(c-a)x+\frac{1}{4}(3a+2b-c)$ ... (i)

following (i) and (ii) we get $(a+c-2b)\left(x-\frac{1}{2}\right)^2=0$

$\Rightarrow$ Line touches curve. $\Rightarrow x=\frac{1}{2}$

Q43

$\int_1^{xy} f(t)dt=y\int_1^x f(t)dt+x\int_1^y f(t)dt$

Differentiate w.r.t. x keeping y as constant

$\Rightarrow f(xy)\cdot y=yf(x)+\int_1^y f(t)dt$

put $x=1$ then

$\Rightarrow y\cdot f(y)=y(1)+\int_1^y f(t)dt$

differentiate w.r.t. y

$y\cdot f'(y)+f(y)=3+f(y)\Rightarrow f'(y)=\frac{3}{y}$

$\Rightarrow f(x)=3\ell nx+C$

$\because f(1)=3\Rightarrow C=3$

$f(x)=3\ell nx+3$

$\int_1^e f(x)\,dx=\int_1^e(3\ell nx+3)\,dx$

$=3(x\ell n-x)^e+3(e-1)$

$=3\left[(e\cdot\ell ne-e)-(0-1)\right]+3(e-1)$

$=3\left[(e-e)+1\right]+3e-3$

$=3e$

$\int_1^e f'(x)\,dx=\int_1^e\frac{3}{x}dx=3(\ell nx)_1^e=3(\ell ne-\ell n1)=3$

Q44

Let $P_1\left(t_1,t^3{}_1\right)$ be a point on $y=x^3$

$\left.\frac{dy}{dx}\right|_{P_1}=3t_1^2$

Tangent at $P_1: y-t^3{}_1=3t_1{}^2(x-t_1)$ .... (i)

solve equation (i) with $y=x^3$

$\mathrm{x}^3-\mathrm{t_1}^3=3\mathrm{t_1}^2(\mathrm{x}-\mathrm{t_1})$

$\Rightarrow (x-t_1)^2(x+2t_1)=0$

If $\mathrm{P}_2\left(\mathrm{t}_2, \mathrm{t}_2{}^3\right)$ then $\mathrm{t}_2=-2\mathrm{t}_1 \quad (\therefore \mathrm{t}_1 \neq \mathrm{t}_2)$

Similarly, the tangent at P_2 will meet the curve at $P_3\left(t_3, t_3{}^3\right)$

$t_3=-2t_2=4t_1$

and so on

$\therefore$ absicssae of $P_1, P_2, \ldots\ldots\ldots\ldots\ldots\ldots P_n$ are

which is G.P.

ordinates of $P_1, P_2, \ldots\ldots\ldots\ldots\ldots P_n$ are

$\mathrm{t}_1{}^3, (-2\mathrm{t}_1)^3, (4\mathrm{t}_1)^3, \ldots\ldots \left((-2)^{\mathrm{n}-1}\mathrm{t}_1\right)^3$

$\Rightarrow \mathrm{t}_1{}^3, -8\mathrm{t}_1{}^3, 64\mathrm{t}_1{}^3, \ldots\ldots (-8)^{\mathrm{n}-1}\mathrm{t}_1{}^3$

which is G.P.

$\therefore \operatorname{Lim}_{n\to\infty} \sum_{r=1}^{n} \frac{1}{x_r} = \frac{1}{1} + \left(\frac{1}{-2}\right) + \frac{1}{4} + \ldots\ldots\ldots \infty$

$= \frac{1}{1-\left(-\frac{1}{2}\right)} = \frac{2}{3}$

Area of $\Delta P_1P_2P_3 = \frac{1}{2}\begin{vmatrix} t_1 & t_1{}^3 & 1 \\ \mathrm{t}_2 & \mathrm{t}_2{}^3 & 1 \\ \mathrm{t}_3 & \mathrm{t}_3{}^3 & 1 \end{vmatrix}$

Area of $\Delta P_2P_3P_4 = \frac{1}{2}\begin{vmatrix} \mathrm{t}_2 & \mathrm{t}_2{}^3 & 1 \\ \mathrm{t}_3 & \mathrm{t}_3{}^3 & 1 \\ \mathrm{t}_4 & \mathrm{t}_4 & 1 \end{vmatrix} = \frac{1}{2}\begin{vmatrix} -2\mathrm{t}_2 & (-2\mathrm{t}1)^3 & 1 \\ -2\mathrm{t}_2 & (-2\mathrm{t}_2)^3 & 1 \\ (-2\mathrm{t}_3) & (-2\mathrm{t}_3)^3 & 1 \end{vmatrix}$

$= 16$ (Area of $\Delta P_1P_2P_3$)

Q45

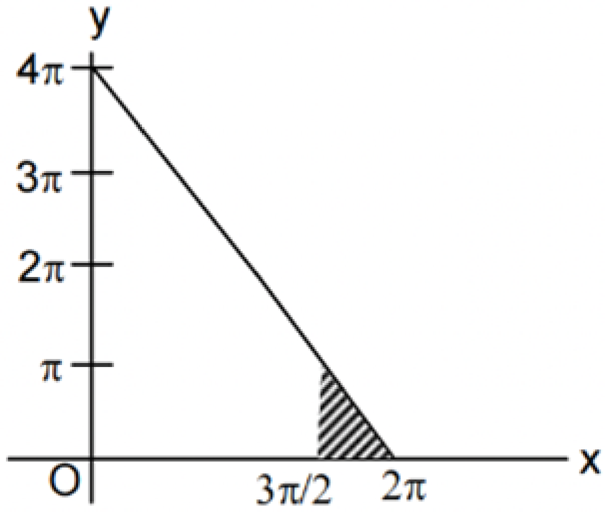
We have, $y=\left|\frac{\pi}{2}-\sin^{-1}(\sin x)\right|-\left|\frac{\pi}{2}-\cos^{-1}(\cos x)\right|$

$=\left|\frac{\pi}{2}-(x-2\pi)\right|-\left|\frac{\pi}{2}-(2\pi-x)\right|=\left|\frac{5\pi}{2}-x\right|-\left|x-\frac{3\pi}{2}\right|$

$=\left(\frac{5\pi}{2}-x\right)-\left(x-\frac{3\pi}{2}\right)=4\pi-2x$

$\left(\because x \in \left[\frac{3\pi}{2}, 2\pi\right]\right)$

Clearly required Area = Area of shaded Δ



$= \frac{1}{2} \times \left(\frac{\pi}{2}\right) \times \pi = \frac{\pi^2}{4} \Rightarrow k = 4$

Q46

$r_3 - r = 4R\sin^2\frac{C}{2} \Rightarrow 10 = 4 \times 5\sin^2\frac{C}{2} \Rightarrow \angle C = \frac{\pi}{2}$

Also $r_3 = s\tan\frac{c}{2} \quad \Rightarrow s = 12 \Rightarrow \Delta = sr = 24$

Now $y = (x-2)(x-8)(x-6)(x-4) + 16$

$= \left(x^2 - 10x + 16\right)\left(x^2 - 10x + 24\right) + 16$

$= (\mathrm{t}+16)(\mathrm{t}+24) + 16, \mathrm{t} = \mathrm{x}^2 - 10\mathrm{x} = (\mathrm{x}-5)^2 - 25, \quad \mathrm{t} \geq -25$

$= (\mathrm{t}+20)^2$

$\mathrm{y}_{\min} = 0 = \mathrm{m}$

$\mathrm{D} + \mathrm{m} = 24 + 0 = 24$

Q47

(A)Let $f(x) = (\cos\theta + x\sin\theta)^n - \cos n\theta - x\sin\theta$ and $x^2 + 1 = (x+i)(x-i)$

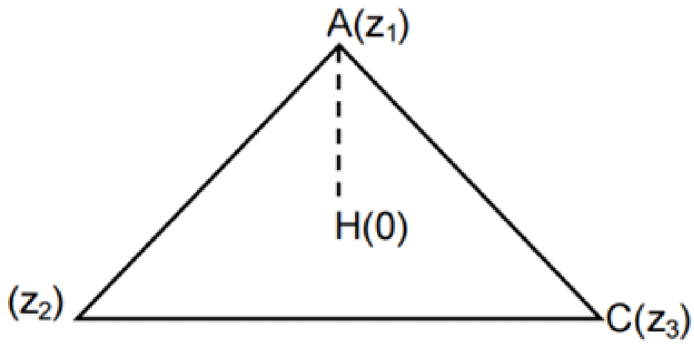
now $f(i) = 0$ and $f(-i) = 0 \Rightarrow f(x)$ is divisible by $x^2 + 1$

(B)Let $f(x) = x^n\sin\theta - k^{n-1}x\sin\theta + k^n\sin(n-1)\theta$

and $x^2 - 2kx\cos\theta + k^2 = 0 \Rightarrow x = k[\cos\theta \pm \sin\theta]$

Now $f[k(\cos\theta + \sin\theta)] = 0 \ \& f[k(\cos\theta - i\sin\theta)] = 0$ (using Demoivre's theorem)

(C)Clearly the angle between BC and AH is $\frac{\pi}{2}$.



$\frac{z_3-z_2}{z_1}$ is purely imaging

$$\frac{z_3-z_2}{z_1}+\frac{\bar{z}_3-\bar{z}_2}{\bar{z}_1}=0$$

$\Rightarrow z_3\bar{z}_1+z_1\bar{z}_3=z_1\bar{z}_2+z_2\bar{z}_1$

$\Rightarrow z_3\bar{z}_1+z_1\bar{z}_3=z_1\bar{z}_2+z_2\bar{z}$

Similarly $BH\perp AC\Rightarrow\frac{z_3-z_1}{z_2}$ is purely imaginary

$\frac{z_3-z_1}{z_2}+\frac{\bar{z}_3-\bar{z}_1}{\bar{z}_2}=0$

$z_2\bar{z}_3+\bar{z}_3z_2=\bar{z}_1z_2+z_1\bar{z}_2$

Q48

W $\rightarrow$ white balls

B $\rightarrow$ back balls

$p(3\text{ W},7\text{ B})\frac{{}^3\text{C}_3\cdot{}^7\text{C}_3}{{}^{10}\text{C}_6}\Rightarrow\text{p}(4\text{ W},6\text{ B})\frac{{}^4\text{C}_3\cdot{}^6\text{C}_3}{{}^{10}\text{C}_6}$

$p(5\text{ W},5\text{ B})\frac{{}^5\text{C}_3\cdot{}^5\text{C}_3}{{}^{10}\text{C}_6}\Rightarrow\text{p}(6\text{ W},4\text{ B})\frac{{}^6\text{C}_3{}^4\text{C}_3}{{}^{10}\text{C}_6}$

$\text{p}(7\text{ W},3\text{ B})\frac{{}^7\text{C}_3\cdot{}^3\text{C}_3}{{}^{10}\text{C}_6}$

$\text{P}\left(\frac{\text{inbag 5 balls white, 5 Black}}{\text{in6 drainball sare white 3 areblack}}\right)$

$=\frac{\left({}^5\text{C}_3\right)^2}{{}^{2.7}\text{C}_3+{}^{2.6}\text{C}_3\cdot{}^4\text{C}_3+\left({}^5\text{C}_3\right)^2}=\frac{100}{70+160+100}$

$=\frac{100}{330}=\frac{10}{33}$

$=\frac{\left({}^5C_3\right)^2}{{}^{2.7}C_3+{}^{2.6}C_3\cdot{}^4C_3+\left({}^5C_3\right)^2}=\frac{100}{70+160+100}$

$=\frac{100}{330}=\frac{10}{33}$

Q49

$x^4 - 12x^3 + bx^2 + cx + 81 = 0$

$\because$ sum of roots = 12 ($\because$ AM = GM)

Product of roots = 81 & All roots positive $\Rightarrow$ roots are 3, 3, 3, 3

$\therefore b = 54\ c = -108$

$2b + c = 0 \Rightarrow x = 1$ is root of $2bx + c = 0$

$\therefore \alpha = 1$

$\therefore f(x) = x^3 + 3x + 1 \Rightarrow f'(x) = 3x^2 + 3$

$g(f(x)) = x \Rightarrow g'(f(x)) \cdot f'(x) = 1$

$\Rightarrow\ f(x) = 5 \Rightarrow\ x^3 + 3x - 4 = 0\ \Rightarrow g'(f(1)) \cdot f'(1) = 1$

$\Rightarrow x = 1 \Rightarrow g'(5) = \frac{1}{6} \Rightarrow 6 \cdot g'(5) = 1$

Q50

$\because \quad f(x) = ((x+1)(x+2))^{\cos_\pi x}$

If x is even integer the $f(x) = (x+1)(x+2)$

If x is odd integer then $f(x) = \frac{1}{(x+1)(x+2)}$

$\left|\sum_{k=1}^{n} \log_{10} f(k)\right| = 1 \Rightarrow \sum_{k=1}^{n} \log_{10} f(k) = \pm 1$

$\Rightarrow \log_{10} f(1) + \log_{10} f(2) + \ldots\ldots + \log_{10} f(n) = \pm 1$

$\Rightarrow f(1) \cdot f(2) \ldots\ldots. f(n) = 10$ or $\frac{1}{10}$ ... (1)

If n is even integer, then from ... (2)

$\frac{1}{(1+1)(1+2)} \cdot (2+1)(2+2) \frac{1}{(3+1)(3+2)} \ldots\ldots (n+1)$

$(n+2) = 10, \frac{1}{10}$

$\Rightarrow \frac{n+2}{2} = 10, \frac{1}{10} \Rightarrow n + 2 = 20, \frac{1}{5}$

$\Rightarrow n = 18 \quad \because n \in N$

If n is odd then from ... (1)

$\frac{1}{(1+1)(1+2)}(2+1)(2+2) \ldots\ldots \frac{1}{(n+1)(n+2)} = 10,\ \frac{1}{10}$

$\Rightarrow \frac{1}{2(n+2)} = \frac{1}{10} \Rightarrow n = 3$

Q51

Here, $2^m + 2^n + 2^p = (3-1)^m + (3-1)^n + (3-1)^p$

$= 3\mathrm{k} + (-1)^{\mathrm{m}} + (-1)^{\mathrm{n}} + (-1)^{\mathrm{p}} \quad (\mathrm{k} \in \mathrm{l})$

So that $2^m + 2^n + 2^p$ is divisible by 3 if m, n, p all are odd or all are even

$\Rightarrow$ Number of possible ordered triplets

$= 50 \times 25 \times 12 + 50 \times 25 \times 13 = 31250 = 6250 \times 5$

Q52

Orthocentre is foot of perpendicular drawn from origin on the plane

$\frac{x}{6} = \frac{y}{3} = \frac{z}{2} = \frac{-(-6)}{36+9+4} \Rightarrow (x, y, z) \equiv \left(\frac{36}{49}, \frac{18}{49}, \frac{12}{49}\right)$

d = 6/7

7d = 6

Q53

$\left(1+x+x^2\right)^{3n+1} = a_0 + a_1 x + a_2 x^2 + \ldots\ldots a_{6n} x^{6n} + a_{6n+1} x^{6n+1} + a_{6n+2} x^{6n+2}$

putting $x = \omega$ and ω^2, we get

$\left(1+\omega+\omega^2\right)^{3n+1} = 0 = a_0 + a_1\omega + a_2\omega^2 + a_3 + \ldots . + a_{6n} + a_{6n+1}\omega + a_{6n+2}\omega^2$... (1)

$\left(1+\omega^2+\omega^4\right)^{3n+1} = 0 = a_0 + a_1\omega^2 + a_2\omega + a_3 + \ldots\ldots + a_{6n+1}\omega^2 + \mathrm{a}_{6\mathrm{n}+2}\omega$... (2)

Adding (1) and (2), we get

$\sum_{r=0}^{2n}\left(2a_{3r} - \left(a_{3r+1} + a_{3r+2}\right)\right) = 0$

Q54

$\frac{c+a}{12} = \frac{a+b}{14} = \frac{b+c}{18} = \frac{s}{11}$

$b + c = \frac{18}{11} s$

$\Rightarrow \quad 2s - a = \frac{18}{11} s$

$\Rightarrow \quad \mathrm{s} - \mathrm{a} = \frac{7}{11}\ \mathrm{s}$

$\mathrm{r}_1 = \frac{\Delta}{\mathrm{s}-\mathrm{a}} = \frac{11\Delta}{7\ \mathrm{s}} = \frac{11}{7} r$

$a + c = \frac{12s}{11}$

$2s - b = \frac{12s}{11}$

$s-b=\frac{s}{11}$

$r_2=\frac{\Delta}{s-b}=\frac{11\Delta}{s}=11\mathrm{r}$

$\Rightarrow \quad \frac{7r_1-r_2}{11}=\frac{11r-11r}{11r}=0$